

PRAWIN KUMAR N

Software Development Engineer | Full Stack Developer | MERN Stack Engineer | Backend Engineer

+91 8807054164 | prawinkumar2k4@gmail.com | linkedin.com/in/prawinkumar-n | github.com/prawinkumar2k | prawin.vercel.app 1,
Kumaran Nagar, Perundurai, Tamil Nadu – 638052 | Open to: Bangalore | Chennai | Remote

PROFESSIONAL SUMMARY

Full Stack Developer and Software Development Engineer with 2 years of hands-on experience building and deploying production-grade government portals, enterprise ERP systems, SaaS platforms, and IoT solutions — while pursuing B.Tech in AI and Data Science. Sole developer of the Tamil Nadu Government DOTE Admissions Portal (tnpoly.in) serving 50,000+ students with 99.9% uptime — complete ownership from architecture to live server deployment. Co-founded a registered Private Limited company and secured external investor funding to build SaaS ERP products for the education sector. Led the ERP development team at Nandha Engineering College. 3x Innovation Day 1st Prize winner. Targeting SDE-1, Full Stack Developer, and Backend Engineer roles.

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), Python, PHP, C, SQL
MERN Stack:	MongoDB, Express.js, React.js 18, Node.js, Vite, Tailwind CSS, Bootstrap, HTML5, CSS3
Backend:	Node.js, Express.js, REST API Design, JWT Authentication, RBAC, Sequelize ORM, Socket.IO, Query Optimisation
Databases:	MySQL, MongoDB (Certified), SQL Server, PostgreSQL, Schema Design, Indexing, Performance Tuning, Aggregation Pipelines
Frontend:	React.js 18, Vite, Tailwind CSS, Bootstrap, HTML5, CSS3, JavaScript DOM, Responsive Design
DevOps/Infra:	Linux (Ubuntu), Nginx, PM2, UFW Firewall, SSL/TLS (Certbot), Docker, Hostinger VPS, GitHub Actions, Azure Cloud, CI/CD
AI/ML/IoT:	Deep Learning, Computer Vision (OpenCV), Machine Learning, TensorFlow, Arduino, ESP32, RFID, Socket.IO
Tools:	Git, GitHub, VS Code, Postman, Tableau, Blender

WORK EXPERIENCE

Software Development Engineer

SearchFirst Technologies Pvt. Ltd., Erode, Tamil Nadu | Jan 2025 – Jun 2026 | Part-time during semester; Full-time during academic breaks

- Co-founded a government-registered Private Limited IT company focused on SaaS ERP and government web applications — secured external investor funding for product development.
- Sole developer of Tamil Nadu Govt DOTE Admissions Portal (tnpoly.in) — Node.js, Express.js, MySQL REST APIs serving 50,000+ students; achieved 40% latency reduction and 99.9% uptime.
- Performed full server lifecycle management: Ubuntu OS installation, Nginx reverse proxy, PM2, UFW firewall, SSL/TLS certificates, and high-level server alert systems.
- Engineered 3-tier RBAC with server-side gender and hostel-based college filtering — cut admin workload by 60% and eliminated 100% of manual eligibility screening.
- Integrated CCAvenue payment gateway across 100+ institutions — fully digital fee collection.
- Optimised RESTful APIs and database schemas across 5+ production applications — consistent 35–40% improvement in query execution time.
- Developed investor-funded SaaS ERP modules for pharmacy colleges, schools, and engineering colleges — covering academics, examinations, HR/payroll, LMS, NAAC/PCI compliance, and finance.

Software Developer — ERP Development Team Lead

Nandha Engineering College, Erode, Tamil Nadu | Jan 2025 – Jan 2026 | 1-Year Certified Commercial Engagement

- Led development and maintenance of Campus ERP (PHP, SQL Server, JavaScript, Bootstrap) serving 1,000+ students — attendance, academics, fee management, and reporting modules.
- Managed team of junior developers — mentored on full-stack architecture, database design, and deployment best practices.
- Received 1-year certified commercial engagement from the institution for software development services.

Web Development with AI Intern

SHENai Private Software Solutions, Erode | Jun 2025 – Jul 2025

- Built and deployed AI-integrated web applications; contributed to production releases within tight deadlines.

UI/UX Design Intern

SHENai Private Software Solutions, Erode | Aug 2024 – Sep 2024

- Designed and prototyped 3+ client-facing UI modules — improved UI responsiveness by 30%; gathered client requirements and shipped iterative releases.

KEY PROJECTS

DOTE Admissions Portal — Tamil Nadu Govt [LIVE | 50,000+ Users]

React 18, Vite, Node.js, Express.js, MySQL, Sequelize ORM, Nginx, PM2, CCAvenue, Ubuntu VPS | Jan 2025 – Present

- Production government portal — 40% latency reduction, 99.9% uptime, 3-tier RBAC, digital payments for 100+ institutions.
- Full server ownership: OS install ☐ Nginx config ☐ SSL/TLS ☐ UFW firewall ☐ PM2 ☐ alert monitoring.

Campus ERP — Nandha Engineering College [LIVE | 1,000+ Students]

PHP, SQL Server, JavaScript, Bootstrap | Jan 2025 – Present

- Institutional ERP for attendance, academics, and fee management — 50% reduction in administrative effort; 1-year certified commercial deployment.

SaaS ERP Platform — Education Sector (Investor-Funded)

MERN Stack, Node.js, Express.js, MongoDB, React.js, MySQL, Tailwind CSS, JWT, Docker | 2025 – Present

- Multi-tenant SaaS ERP with modules for academic management, examination automation, HR/payroll, LMS, library, NAAC/PCI compliance, and finance — built under funded product development.

Enterprise Payroll & HRMS Platform [Production]

Node.js, Express.js, MongoDB, React.js, Tailwind CSS, JWT, RBAC | 2025

- 3-role HRMS (Admin, HR, Employee) with salary processing APIs for 100+ employees — MongoDB schema restructuring cut query time by 35%.

Hospital Management System [Production]

React.js, Node.js, Express.js, MySQL | 2025

- Patient records, appointments, billing — reduced manual data entry by 70% with role-based clinical dashboards.

AI-Based Autonomous Smart Car [1st Prize — Innovation Day '24 & '25]

Arduino, ESP32, RFID, Deep Learning (CNN), OpenCV, Python | 2024–2025

- CNN-based path navigation, RFID security, AI voice assistant — 1st Prize at Nandha Innovation Day 2024 and 2025 (two consecutive years).

Smart RFID Attendance System

Arduino, RFID, PHP, SQL Server | 2024

- Automated RFID attendance for 200+ students — eliminated 100% manual recording.

EDUCATION

B.Tech – Artificial Intelligence and Data Science

Nandha Engineering College, Erode (Anna University Affiliated, NAAC A+) | 2023 – 2027 (Final Year) | CGPA: 7.1 (Sem 1–5)

Coursework: Deep Learning, Computer Vision, ML, IoT, Database Design, OS, Data Science, AR/VR, Advanced Coding, Tableau

HSC (12th) — Shri Ganga Higher Secondary School, Ingur | Mar 2023 | 405/600

SSLC (10th) — Shri Ganga Higher Secondary School, Ingur | Mar 2021 | Pass

CERTIFICATIONS

- MongoDB Certified Associate Developer — MongoDB, Inc. | Mar 2025 | No Expiry
- Python — GUVI / Google for Education Partner | Aug 2023
- AI for India 2.0 — GUVI / Skill India Digital / NSDC | Aug 2023
- Internet of Things (IoT) Workshop — NoviTech R&D; Pvt. Ltd. (ISO 9001:2015) | Sep 2023
- Industrial Visit — SinroRobotics Pvt. Ltd. (ISO Certified, QCI) | Sep 2024
- Python Programming Quiz — CodeAlpha | Sep 2024

ACHIEVEMENTS & AWARDS

- ★ 1st Prize — AI-Based Autonomous Car | Nandha Innovation Day 2025 (Edition 14) | Feb 2025
- ★ 1st Prize — AI-Based Autonomous Car | Nandha Innovation Day 2024 (Edition 13) | Feb 2024 (Two consecutive years)
- ★ 1st Prize — Image Caption Project | Nandha Innovation Day 2025 (Edition 14) | Feb 2025
- ★ Participated — Automated Pani Puri Dispensing Machine | Nandha Innovation Day 2026 (Edition 15) | Feb 2026
- ★ 1st Prize — Hack Elite 2K23, CyberVault Event | Excel Engineering College, Dept. of IT | Nov 2023
- ★ Best Student in Co-Curricular Activities | Nandha Engineering College Annual Day | Dept. of AI | Mar 2024
- ★ Co-Founded a registered Private Limited IT company — secured INR 25 Lakh external investor funding | Aug 2025
- ★ Delivered Tamil Nadu Govt DOTE Portal (tnpoly.in) — 50,000+ users, zero critical downtime, full server ownership
- ★ Horizon'25 National Level AI Hackathon | KPR Institute / ACM Student Chapter | Apr 2025
- ★ Yantrik'23 National Technical Symposium | KPR Institute of Engineering | Oct 2023
- ★ Voluntary Blood Donation — Tamil Nadu State AIDS Control Society | Oct 2024
- ★ 37+ GitHub repositories | 296 contributions in the last year

ADDITIONAL

Languages: Tamil (Native), English (Professional) Passport: Valid | Available for relocation DOB: 07-Dec-2004